Before anything else: your dataset is unique to you

This assessment uses **real, published data**, but the particular slice of it that you analyse is derived from your student ID. No two students in this unit receive the same countries, the same range of years, or the same data-quality problems.

Every number you report; row counts, totals, top-10 tables, correlations, model errors, is specific to your dataset. Three things follow from that:

1. **An answer copied from another student will be numerically wrong**, because it was computed on a different slice of the data.
2. **An answer produced without actually running your code will be wrong**, for the same reason.
3. **Your marker regenerates your dataset from your student ID and checks your reported values automatically**, including the numbers you read aloud in your video. Unexplained mismatches are referred under the Academic Integrity Policy.

The underlying figures are genuine. Nothing has been simulated or randomised in the *values*, so you can sanity-check any country against the published source. What differs between students is *which* countries and years you hold, and *which* defects are present in your copy of the files.

What you've been given

Download and unzip `FinalAssessment_2026.zip`. Inside you'll find:

File	What it is
`Submission_Template.Rmd`	**This is the file you work in and submit.** It has a section for every task, ready for your code and your writing.
`personalise.R`	Builds your personal copy of the data. Don't edit it.
`data_raw/`	The shared source files. Don't edit these either.

This brief is your reference, so please keep it open beside the template.

Getting started

Open `Submission_Template.Rmd` in RStudio and find the YAML header at the top. Replace `uXXXXXXX` with your real student ID:

```
params:
  studentID: "u1234567"
```

Then run the first code chunk. It reads the shared files from `data_raw/` and writes **your** four CSV files into a new folder called `data/`. Every task in this assessment uses the files in `data/`.

The chunk prints a **fingerprint**; a code like `EM-681782-8637-8645`. Write it down. You report it in Section 5 and state it in your video, and it is how your marker confirms which dataset your results came

from.

> **Do not edit `personalise.R`.** Your marker regenerates your data from a pristine copy. If yours has been changed, your fingerprint won't match and your submission can't be verified.

If you leave `uXXXXXXX` in the header by mistake, no harm done, put your real ID in, delete the `data/` folder, and run the chunk again.

Assignment instructions

Type. Take-home assignment.

Weighting. 100 marks.

Structure. Four tasks. Each requires **R code** together with **written description and interpretation of your own results**.

What you submit. A single ZIP file named `[studentID]_final_assessment.zip` containing:

1. Your completed `Submission_Template.Rmd`, with your student ID in the YAML header and `personalise.R` unmodified.
2. The **PDF/HTML** produced by knitting that exact `.Rmd`.
3. Your **hand-drawn dashboard sketch**, embedded as an image inside Task 4.
4. Your **Decision Log**, completed in the appendix of the template (minimum 5 entries).
5. Your **video**, see below more instructions about this.

Where. The unit's Canvas site, by **10:00, Tuesday of Week 14**.

Coverage. This assessment evaluates all learning outcomes and integrates content taught throughout the unit.

Marking criteria. See the rubric on Canvas. Marks include the overall presentation of each task, including coding style — see Google's R Style Guide.

The video walkthrough

A **5–8 minute recorded video** is submitted alongside your written work. You'll demonstrate your own code running live and talk through what it does. It is assessed on completion and authenticity, not on polish; you don't need to be scripted. Full instructions are in the `Video_Walkthrough_Instructions` file on Canvas. Read them early.

The Decision Log

Keep a running **Decision Log** in the appendix of your template. Record at least **five non-trivial analytical decisions** — how you handled a particular anomaly, why you chose a specific chart type, why you joined two files one way rather than another.

For each entry, state *(a)* the decision, *(b)* the alternative you considered, and *(c)* why you chose one over the other.

This log must reflect your **actual process as it happened**, not a summary written at the end. Write each entry at the time you make the decision. Generic entries ("I chose the best approach") receive zero marks. Every entry must point at something specific in your own work: a variable name, a country, a line of code, an output value.

The data

You are given four CSV files describing **national carbon dioxide emissions and electricity generation** over roughly two decades. They come from Our World in Data's open CO_2 and energy datasets, which compile figures from the Global Carbon Project, Ember, the Energy Institute and the World Bank.

The files are messy. They arrive in inconsistent shapes and formats, they do not share a single join key, their coverage is uneven, and they contain data-quality defects of several different kinds. Finding those defects, deciding what to do about each one, and defending that decision is a substantial part of what is being assessed.

You are not told how many defects there are, or where. That is the task.

File descriptions

`data/Emissions.csv` — the master panel

One row per country per year.

Column	Description
`iso_code`	ISO 3166-1 alpha-3 country code
`location`	Country name
`region`	Continent
`year`	Year of observation
`annual_co2_mt`	Annual production-based CO_2 emissions (million tonnes)
`annual_oil_co2_mt`	Annual CO_2 emissions from oil (million tonnes)

`data/Renewables.csv` — renewable electricity generation

In **wide** format: one row per country, one column per year, values in terawatt-hours (TWh).

`data/Consumption.csv` — consumption-based emissions

Covers only a subset of countries.

Column	Description
`Country Code`	ISO alpha-3 code
`Year`	Year of observation

Column	Description
`Consumption CO2`	Annual consumption-based CO_2 emissions (million tonnes)

`data/Countries.csv` — static country reference

Column	Description
`countryCode`	ISO alpha-3 code
`Country`	Country name
`popData2022`	Population, 2022
`GDP`	Gross domestic product (millions of international dollars)
`GDP/capita`	GDP per person (international dollars)

Production-based vs consumption-based emissions

This distinction matters for Tasks 2 and 3, so make sure you understand it before you start.

Production-based emissions (`annual_co2_mt`) are emitted inside a country's borders.

Consumption-based emissions (`Consumption CO2`) are adjusted for trade: emissions embodied in imported goods are added, and those embodied in exported goods are subtracted.

A country that imports manufactured goods has consumption-based emissions *higher* than its production-based emissions. A manufacturing exporter has the reverse.

Source and copyright

1. OWID CO_2 and Greenhouse Gas Emissions — https://github.com/owid/co2-data
2. OWID Energy dataset — https://github.com/owid/energy-data
3. Global Carbon Project — https://globalcarbonbudget.org
4. Ember — https://ember-energy.org

Task 1: Data preparation and wrangling

1. Import the data from the four CSV files in `data/` and store them in appropriately named data frames.

2. Tidy the data frame derived from `Renewables.csv` so that it is compatible with the one derived from `Emissions.csv` — that is, every observation should be one country in one year. Note that this file contains rows that are **not countries**. Deal with them, and say what you did.

3. Rename columns as follows.

File	Original	Rename to
`Emissions.csv`	`iso_code`, `location`, `region`, `year`, `annual_co2_mt`, `annual_oil_co2_mt`	`ISO`, `Country`, `Region`, `Year`, `CO2`, `OilCO2`
`Consumption.csv`	`Country Code`, `Year`, `Consumption CO2`	`ISO`, `Year`, `ConsCO2`
`Countries.csv`	`countryCode`, `Country`, `popData2022`, `GDP`, `GDP/capita`	`ISO`, `Country`, `Population`, `GDP`, `GDPCapita`

4. Ensure `Year` is stored consistently, same type, same format, in every data frame. At least one file will not cooperate. *Hint: `lubridate` may help with some of the variants you encounter.*

5. Taking the `Emissions.csv` data frame as the master, add **five new variables** to it from the other three files: `Renewables`, `ConsCO2`, `Population`, `GDP`, `GDPCapita`.

   The files do not all join on the same key. Report how many rows failed to match on each join, and what you did about them.

6. Check for NAs in the merged data frame and replace them with quantities of your choosing. Justify each choice; different variables may warrant different treatments.

7. Using the existing `Year` variable, add two new variables: `Decade` (e.g. `2000s`, `2010s`, `2020s`) and `Period`, taking the value `"Pre-Paris"` for years before 2016 and `"Post-Paris"` from 2016 onwards.

8. Add four new variables, `CumCO2`, `CumOilCO2`, `CumRenew`, `CumConsCO2`, reflecting the cumulative value for that country up to and including the year of the observation. For example, `CumCO2` for country "X" in year "Y" is the total emitted by "X" from the first year in the record up to "Y".

9. Add two new variables:

   - `NetImported` = `CumConsCO2 - CumCO2`. A positive value means the country consumes more emissions than it produces — it is a net importer of embodied carbon.
   - `OilShare` = `CumOilCO2 / CumCO2`, the proportion of cumulative emissions from oil.

10. Add four new variables — `CO2_PC`, `OilCO2_PC`, `Renew_PC`, `ConsCO2_PC` — reflecting the cumulative quantity **per person**, in tonnes (or TWh per million people for renewables). *Hint: `CO2_PC = CumCO2 * (10^6) / Population`, since emissions are recorded in million tonnes.*

1A. Data Quality Audit Report

After completing the wrangling above, write a structured **Data Quality Audit** of approximately **250 words**. It must reference specific values, country names, years and column names **from your own data**. Generic or templated answers receive no marks.

i. Defect inventory

Your files contain data-quality defects of **several distinct kinds**. You are not told how many or where.

Produce a table listing every defect you found, with these columns: *file*, *column*, *how many rows affected*, *how you detected it*, *how you handled it*. For each defect, name at least one specific country and year.

> Marks are awarded for the **detection method**, not just the finding. "I looked at the data" is not a detection method. "`summary()` showed a minimum that isn't physically possible" is.

ii. The negative emissions problem

At least one row in your `Emissions.csv` records a negative annual emissions figure. Identify these rows by country and year.

A negative annual emissions figure is physically implausible for these countries — but negative values **do** legitimately appear in published emissions series. Explain the circumstance under which that happens. Consider the difference between a *revision to a previously published estimate* and an *actual removal of carbon from the atmosphere*.

Justify the handling decision you made in your code, and explain one alternative strategy you chose **not** to adopt, and why.

iii. Coverage gap in consumption-based data

`Consumption.csv` covers only a subset of the countries in `Emissions.csv`. Compute and report the **exact number** of countries in your `Emissions.csv` that have *no* consumption-based data at all, and list three of them by name.

Then: is replacing the missing values with zero a valid imputation strategy? If it introduces bias, state **in which direction** `ConsCO2_PC` and `NetImported` are biased, and **for which type of country** the bias is most severe.

Support your answer with a comparison of the countries that do and do not have coverage. A claim with no supporting number scores no marks.

Task 2: Exploratory data analysis

1. Find the year with the highest total CO_2 emissions across all countries in your dataset. Print the year and the total. Then find the year with the **largest year-on-year fall** in total emissions, print it, and state in one sentence what you think caused it.

2. Build a graph showing how the cumulative totals of CO_2, oil CO_2, renewables and consumption CO_2 change over time for your whole dataset collectively.

3. Extract the records for the **final year in your dataset** for all countries, and save them in a separate data frame named `lastYear_data`. *Hint: filter on `Year == max(Year)`. Do not hard-code a year — yours may differ from your classmates'.*

4. Based on `lastYear_data`, extract the top 10 countries by cumulative emissions, by cumulative emissions per capita, by net imported emissions, and by renewable generation per capita, into separate data frames named `top10total`, `top10percap`, `top10net` and `top10renew`.

5. Based on `lastYear_data`, print total cumulative CO_2, oil CO_2, renewables and consumption CO_2 for every `Region`.
6. Build a graph showing total cumulative emissions over time for the top 10 countries from question 4. Use a log transformation on the Y axis.
7. Build a graph for the top 10 countries by cumulative emissions, with one sub-graph per country (use `facet`), showing how annual CO_2, annual oil CO_2 and annual renewable generation changed over time. Use different colours to distinguish the variables.
8. Build a graph for the top 10 countries by cumulative emissions per capita from question 4, presenting cumulative emissions, population and emissions per capita for each country.
9. Build a graph presenting the statistics (total, average, median) of cumulative emissions by region. Use the region in the legend, and ensure the x-axis labels do not overlap.
10. Based on `lastYear_data`, list the top 2 countries in each region by cumulative emissions per capita.

2A. Counterintuitive finding

Compare your `top10total` and `top10percap` tables. They will not contain the same countries.

Review all the visualisations and summaries you produced in Task 2, and identify **one finding that contradicts what you would have expected before looking at the data**.

Your response must:

- State explicitly what you *expected*, and why that expectation was reasonable.
- State what the data *actually shows*, with a direct reference to a specific value, country name, or trend visible in one of **your** plots.
- Propose at least one plausible real-world mechanism for the discrepancy — population size, industrial structure, energy resource endowment, trade patterns, or the production-versus-consumption accounting distinction.
- Produce **one additional visualisation of your own design**, not required elsewhere in Task 2, that supports or investigates your explanation. It must have a title, labelled axes, and a 2–3 sentence caption explaining what the chart shows and why you chose that chart type.

2B. Visualisation design critique

Select **one** visualisation you produced in Task 2, and state which question it is from.

Write a structured critique of approximately **150 words** covering:

- **Two design decisions** you made — log scale, choice of geom, colour scheme, faceting. For each, explain what it communicates and why it suits this data.
- **One alternative design** you considered but rejected. Explain concretely why it would be *worse* at communicating the intended message to a climate-policy audience. Do not simply say it "looks worse".
- **One limitation** of your chosen visualisation that a reader could misinterpret. Describe the misinterpretation and how you would mitigate it.

Task 3: Data-driven modelling

1. From your master data frame, create a separate data frame named `cor_data` containing the **final-year** record for each country, with the variables `CumCO2`, `CumConsCO2`, `Population`, `GDP`, `GDPCapita`.
2. Compute the correlation matrix between the variables of `cor_data` and visualise it. Report the correlation between `CumCO2` and `GDP`, and between `CumCO2` and `GDPCapita`. Comment in one sentence on why those two numbers are so different.
3. Visualise the distribution of `CumCO2` with and without a log transformation on the x axis.
4. Divide `cor_data` into training and testing sets, with training representing 65% of the rows.

   > **You must set the seed as follows immediately before splitting**, so that your split is reproducible and your marker can verify your results:
   >
   > ```
   > set.seed(seed_from_id(params$studentID))
   > ```

5. Train a linear regression model predicting `CumCO2` from `GDP` alone. Evaluate it on the test data and print the RMSE.
6. Train a second linear regression model predicting `CumCO2` from all the other variables. Evaluate it on the test data and print the RMSE.
7. Interpret the two models and write a short report on the differences. In which circumstances should the first model be preferred, and in which is the second better?

   > If the model with more predictors has the **higher** test RMSE, that is not an error. Explain it.

8. Extend the analysis by training alternative regression models — for example Ridge, Lasso, or a decision tree — to predict `CumCO2` from all variables. Compare performance (RMSE, R^2) against the linear models in a table.

3A. Assumption checking before modelling

Evaluate the assumptions of linear regression using your **training data**.

i. Linearity. Plot `CumCO2` against each predictor (`CumConsCO2`, `Population`, `GDP`, `GDPCapita`) using scatter plots. Does a linear relationship appear reasonable? Which predictor shows the clearest departure from linearity?

ii. Distribution of the response. Using the plots from question 3, comment on whether raw `CumCO2` satisfies the normality assumption required for valid inference. Does this suggest a transformation is warranted?

iii. Residual diagnostics. After fitting the full model from question 6, produce the four standard diagnostic plots — residuals vs fitted, Q-Q, scale-location, residuals vs leverage. For each, write one

sentence stating *(a)* what assumption it tests and *(b)* whether your model satisfies it.

iv. Influential observations. Your residuals-vs-leverage plot will identify at least one high-leverage country. **Name it.** Refit the full model with that observation removed, report the new test RMSE, and state in two sentences whether removing it is defensible.

v. Conclusion. Is linear regression a valid choice for this data? If assumptions are violated, what is the practical consequence for interpreting coefficients or predictions?

3B. Feature engineering critique

The model in question 5 uses `GDP` as the sole predictor of `CumCO2`. Address the following in approximately **300 words**.

i. Theoretical critique. Unlike many socioeconomic predictors, GDP has a *physically plausible* causal pathway to emissions. State that pathway. Then explain why a high correlation still does not license a causal claim here: identify at least **two confounding or mediating variables** that `GDP` is acting as a proxy for. Reference the correlation value **you** computed in question 2, and contrast it with the `GDPCapita` correlation.

ii. Proposed feature. Propose **one engineered feature** not present in the dataset that would be a stronger and more theoretically justified predictor of `CumCO2`. Describe precisely how it would be constructed — what raw data is needed and how you would calculate it — and explain why it is more causally connected to emissions than GDP.

iii. Ethical risk. Suppose this model is adopted by an international body to allocate climate finance or to set national emissions-reduction obligations. Identify **one specific way** it would produce systematically unfair outcomes, and describe the mechanism through which that unfairness arises **from the structure of the model itself**, not merely from "bias in the data".

Your dataset contains a direct illustration. Compare a country with large positive `NetImported` against one with large negative `NetImported`, name both, and use them in your argument.

Task 4: Insights

You have been asked to design a handcrafted dashboard, on Microsoft Visio or on paper, that visually represents the trends and key figures of the **global energy transition** over the period covered by your data.

Instructions

1. **Data augmentation.** Your dataset covers national emissions and renewable generation. Explain how you would augment it to give a comprehensive view of the transition. What additional data sources would you bring in, and what would each add?
2. **Key insights.** Identify the most interesting insights you want to highlight for a public audience. What figures — line graphs, bar charts, heatmaps, maps — will you include, and why is each important for understanding emissions trends and the transition?

3. **Design elements.** Describe the layout: colour schemes, typography, and the placement of figures to support readability and engagement.

4. **Audience consideration.** Discuss how your design serves a public audience — accessible, understandable, actionable.

What to submit for Task 4

- A **physical or hand-drawn sketch** of your dashboard layout, photographed and embedded in your template as an image. This is **mandatory**; submissions without it lose all design marks. It need not be artistic, but it must show panel placement, chart types and labels.
- Your dashboard's **narrative thesis** in a single sentence, stated before the design: the one overarching story a reader should take away. Every design decision should serve it.
- Narratives for each of the four points above.
- Responses to these three questions:
    - Identify **one metric or visualisation you deliberately omitted** despite its relevance. Justify why omitting it serves clarity.
    - Identify **one way your dashboard could be misread** by a public audience. Describe the misinterpretation and how you would mitigate it through design.
    - Your dashboard reports emissions for countries of very different sizes. State whether your headline figure uses **total** or **per-capita** emissions, and defend that choice against the obvious objection to it.

Section 5: Verification

Your submission template contains a short table in Section 5.1. Fill it in by hand from your own knitted output. Your marker's script regenerates your dataset from your student ID and checks these values automatically.

The thirteen values are:

1. Your dataset fingerprint
2. First and last year in your dataset
3. Number of countries in your `Emissions.csv`
4. `dim()` of your final merged master data frame
5. Number of duplicate rows you removed in Task 1
6. Countries and years of the negative-emissions rows (1A ii)
7. Number of countries with no consumption-based data (1A iii)
8. Year with the highest total CO_2, and that total (Task 2.1)
9. Top country by cumulative emissions, and by emissions per capita
10. `cor(CumCO2, GDP)` and `cor(CumCO2, GDPCapita)`, to 2 dp
11. Test RMSE of the GDP-only model (Task 3.5)
12. Test RMSE of the full model (Task 3.6)
13. The high-leverage country you identified (3A iv)

Leave no cell blank. If you could not compute one, write `NA` and say why.

Section 5.2 of the template contains a short reproducibility chunk. Run it unchanged as the last code chunk of your submission.

Grading

Marks follow the rubric on Canvas. Three rules apply across all tasks.

A note on AI tools

Using an AI tool to help you *understand* R is permitted, such as asking it how `pivot_longer` works, or why your loop won't run.

Submitting reasoning that is not your own is not permitted. This assessment is built so that the difference between the two is visible, and the video walkthrough is where it is most visible.

Checklist before you submit

- ☐ Your real student ID is in the YAML header of the template
- ☐ `personalise.R` is unmodified
- ☐ All four tasks attempted, with code **and** written interpretation
- ☐ Every interpretive claim quotes a number from your own output
- ☐ Hand-drawn dashboard sketch embedded in Task 4
- ☐ Section 5.1 completed — all thirteen values, none blank
- ☐ Section 5.2 reproducibility chunk present and run
- ☐ Decision Log has at least five specific entries
- ☐ The `.Rmd` knits cleanly to PDF from a fresh session
- ☐ ZIP named `[studentID]_final_assessment.zip` contains the `.Rmd` and the PDF
- ☐ Video recorded, 5–8 minutes, uploaded inside the zip file

Good luck.